STATS I · WEEK 3

Stats I week 3 formulas: Describing numerical data

Numerical data: tables & charts · Mean, median & mode · Range, variance & standard deviation · Percentiles, quartiles, IQR & outliers. Save this PDF for revision.

Numerical data: tables & charts

Class width

upper limit − lower limit

Class mark (midpoint)

(lower + upper)/2

Mean of grouped data

$$(\Sigma f_i x_i)/(\Sigma f_i)$$

x_i = class midpoint

NUMERICAL DATA: TABLES & CHARTS · OFTEN ASKED

Key points

- Class [10, 20) includes 10 and excludes 20; a value of exactly 20 goes in the next class.
- Histogram: bars touch (no gaps); bar area ∝ frequency. Bar charts are for categories, histograms for numbers.
- Stem-and-leaf: stem 2 | leaf 3 means 23. Number of leaves = n. You keep every actual value, so counts, median and mean can be read straight off.
- Shape: symmetric (mean ≈ median); right-skewed, long tail to the right (mean > median); left-skewed (mean < median).

NUMERICAL DATA: TABLES & CHARTS · OFTEN ASKED

Common traps

⚠ 'Above 23' excludes 23; 'at least 23' includes it.

MEAN, MEDIAN & MODE · IN EVERY PAPER

Mean, median & mode

Mean

$$\bar{x} = (\Sigma x_i)/n$$

Median (sorted data)

n odd: value at position (n + 1)/2 · n even: average of positions n/2 and n/2 + 1

Mode

most frequent value (there may be none, or several)

Combined mean

$$(n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$$

MEAN, MEDIAN & MODE · IN EVERY PAPER

Mean, median & mode

Add one value x

$$\text{new mean} = (n\bar{x} + x)/(n + 1)$$

Fix a wrong value

$$\text{new mean} = \bar{x} + (\text{correct} - \text{wrong})/n$$

y = ax + b

$$\bar{y} = a\bar{x} + b$$

Median, mode, quartiles and percentiles transform the same way.

MEAN, MEDIAN & MODE · IN EVERY PAPER

Common traps

⚠ Sort before finding the median.

⚠ Outliers pull the mean; the median barely moves. Skewed data ⇒ the median is the better centre.

⚠ Order matters: $2(\bar{x} - 5) \neq 2\bar{x} - 5$.

RANGE, VARIANCE & STANDARD DEVIATION · IN EVERY PAPER

Range, variance & standard deviation

Range

$$\max - \min$$

Population variance

$$\sigma^2 = (\Sigma(x_i - \mu)^2)/n = (\Sigma\, x_i^2)/n - \mu^2$$

Sample variance

$$s^2 = (\Sigma(x_i - \bar{x})^2)/(n - 1) = (\Sigma\, x_i^2 - n\bar{x}^2)/(n - 1)$$

Standard deviation

$$\sqrt{\text{variance}}$$

Same units as the data.

RANGE, VARIANCE & STANDARD DEVIATION · IN EVERY PAPER

Range, variance & standard deviation

Sample ↔ population

$$s^2 = n/(n - 1)\,\sigma^2$$

y = ax + b

$$\mathrm{Var}(y) = a^2\,\mathrm{Var}(x),\ \mathrm{SD}(y) = |a|\,\mathrm{SD}(x)$$

Recover Σx²

sample: $(n - 1)s^2 + n\bar{x}^2$ · population: $n(\sigma^2 + \mu^2)$

RANGE, VARIANCE & STANDARD DEVIATION · IN EVERY PAPER

What changes under y = ax + b

Measure	Adding b	Multiplying by a
Mean, median, mode, quartiles	+ b	× a
Range, IQR	no change	× |a|
Variance	no change	× a^2
Standard deviation	no change	× |a|

RANGE, VARIANCE & STANDARD DEVIATION · IN EVERY PAPER

Common traps

⚠ Sample ⇒ divide by n – 1. Population ⇒ divide by n. Read 'sample' or 'population' in the question.

⚠ Adding or subtracting a constant never changes variance, SD, range or IQR.

⚠ Multiplying by –2: variance × 4, SD × 2 (SD is never negative).

⚠ Round only at the very end; carry at least 4 decimals in between.

PERCENTILES, QUARTILES, IQR & OUTLIERS · IN EVERY PAPER

IITM's percentile method (the answer keys use it)

Step 1

sort the data in ascending order

Step 2: index

$i = P/100 \times n$

Step 3

i not whole ⇒ round up, take that value · i whole ⇒ average the values at positions i and i + 1

Quartiles

$Q_1 = P_{25}$, $Q_2 = P_{50}$ = median, $Q_3 = P_{75}$

PERCENTILES, QUARTILES, IQR & OUTLIERS · IN EVERY PAPER

IITM's percentile method (the answer keys use it)

IQR

$Q_3 - Q_1$

Five-number summary

min, Q_1, median, Q_3, max

This is what a box plot draws.

Outliers

below $Q_1 - 1.5 \times IQR$ or above $Q_3 + 1.5 \times IQR$

Common traps

⚠ Round up even when the decimal is small: 1.25 → 2, not 1.

⚠ Calculators and Excel use different quartile methods. Use the IITM steps above.

⚠ Q_3 separates the top 25% from the bottom 75%.

⚠ Box-plot whiskers end at the most extreme values that aren't outliers; outliers are drawn as dots.

Notes for every week of Stats I, free.

Full notes, the formula sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/stats-1 →

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